

12. BigInt

Kya hai?

Ek special number type jo **kitne bhi bade integers** store kar sakta hai (normal Number ki limit se bahar).

Banane ke tarike

js

```
const a = 1234567890123456789012345678901234567890n; // end me n
const b = BigInt("1234567890123456789012345678901234567890");
const c = BigInt(10); // 10n
```

Math

js

```
1n + 2n; // 3n
5n / 2n; // 2n (decimal hissa hat jaata hai, zero ki taraf round)
```

Sab operations bigint hi return karte hain.

Mix nahi kar sakte

js

```
1n + 2; // Error: Cannot mix BigInt and other types
```

Convert karna padega:

js

```
BigInt(2) + 1n; // number -> bigint
Number(1n) + 2; // bigint -> number
```

Dhyan: bahut bada bigint `Number` me convert karoge to extra bits **cut** ho jaate hain (error nahi aata).

Unary plus `+bigint` **allowed nahi** hai. `Number()` use karo.

Comparison

js

```
2n > 1;      // true (< > chalte hain)
1 == 1n;     // true
1 === 1n;    // false (alag types)
```

Boolean me

- `0n` falsy hai, baaki sab truthy.

js

```
0n || 2;     // 2
1n || 2;     // 1n
```

Polyfill

BigInt ka achha polyfill mushkil hai (operators ka behavior alag hota hai). Isliye **JSBI** library hai: `JSBI.add(a, b)` jaise calls likhte ho, aur Babel plugin ise native bigint me badal deta hai.

Yaad rakho

- Number ke saath mix nahi.
- `n` suffix ya `BigInt()` se banao.
- Bahut bade integers (jaise IDs, cryptography) ke liye useful.